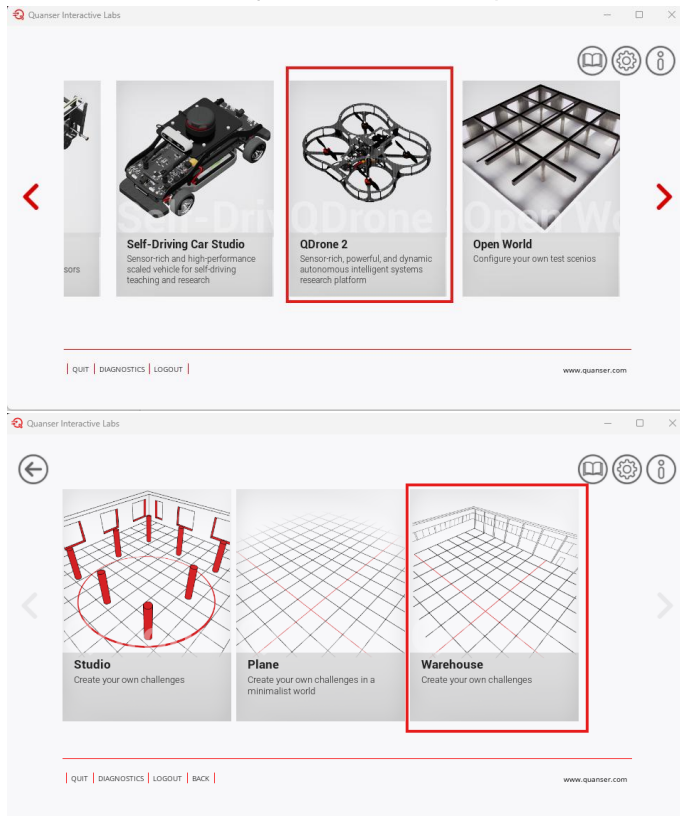
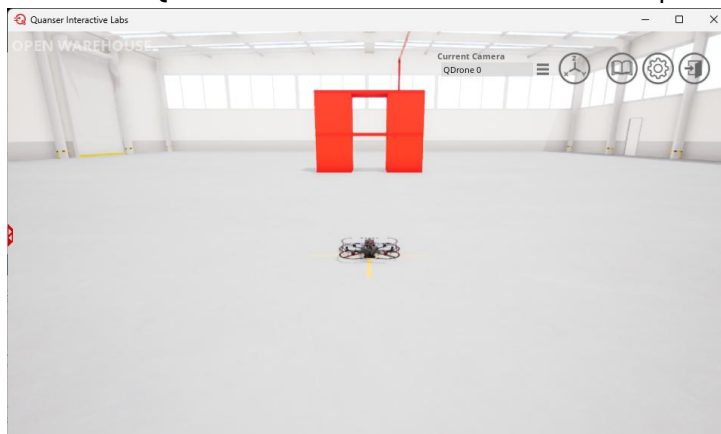


Virtual QDrone 2 Position Control Instructions – For QLabS

1. Open the Quanser Interactive Labs (QLabs) software.
2. Go to the QDrone 2 | Warehouse workspace.

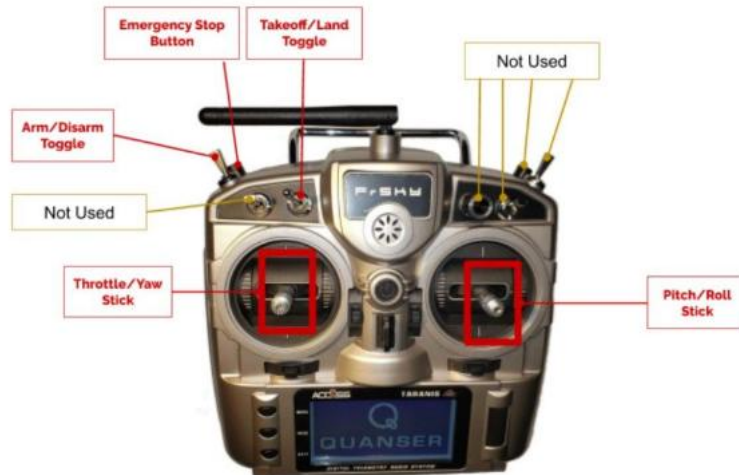


3. Run the **Spawn_QDrone2.m** MATLAB script.
4. The Virtual QDrone 2 should be created in the workspace.



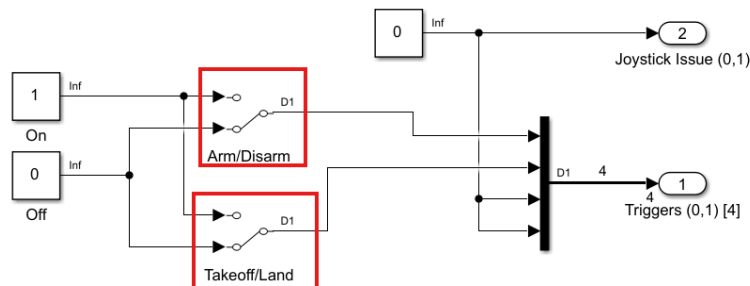
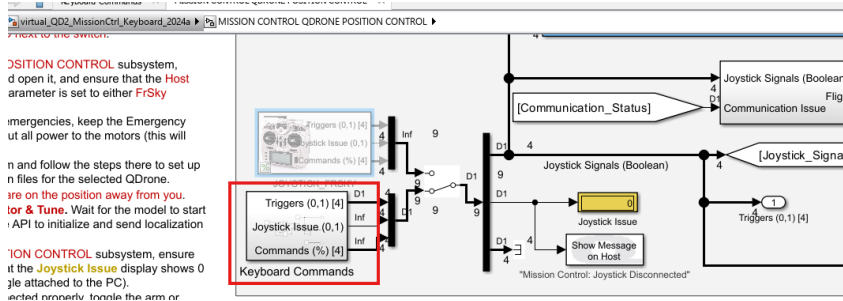
5. Open one of the following:
 - a. **Option 1:** The *virtual_QD2_MissionCtrl_2024a* Simulink model in MATLAB if using the FrSky Simulator joystick.

- b. **Option 2:** The virtual_QD2_MissionCtrl_Keyboard_2024a Simulink model to use the keyboard commands.
6. Click on the **Run** button in Simulink.
7. Open a second MATLAB application.
8. Open the virtual_QD2_DroneStack_PID_2024a_v2 in Simulink in the second MATLAB.
9. Click on the **Run** button in Simulink.
10. **Option 1:** If using the *virtual_QD2_MissionCtrl_2024a* Simulink model, use the joystick toggles to *Arm* and then *Takeoff*.

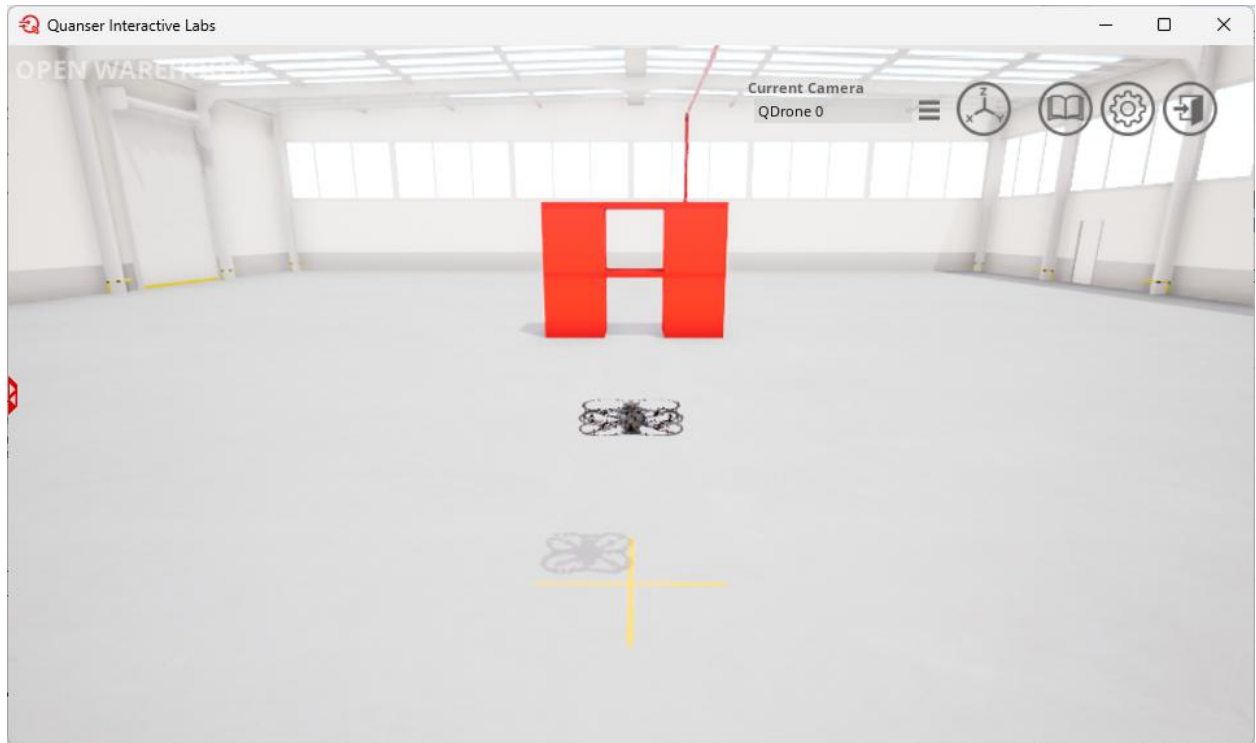


NOTE: Current release of joystick firmware should read "Quanser V2" which is compatible with the mapping in this model

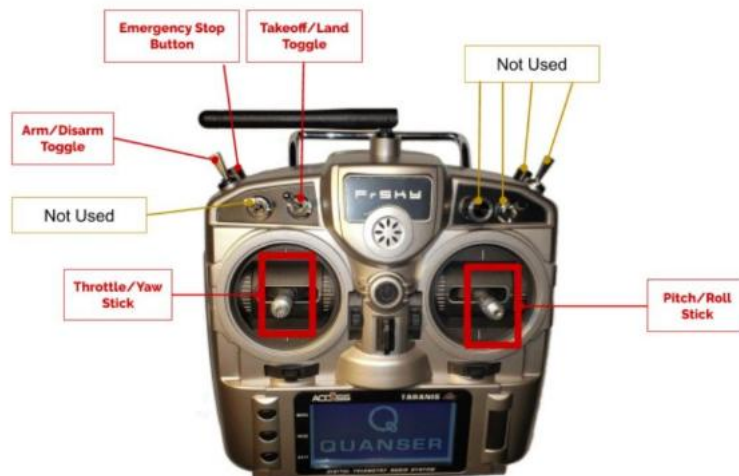
Option 2: If using the virtual_QD2_MissionCtrl_Keyboard_2024a Simulink model, go to the Keyboard Command subsystem to Arm and Takeoff the virtual QDrone 2.



11. The Virtual QDrone 2 should start hovering.



12. Apply position commands by placing the **Command Enable** switch to *Custom Mode* and the **Trajectory Enable** switch to *Enable*.
13. The Virtual QDrone 2 should follow the waypoint commands.
14. When done, set the **Trajectory Enable** switch to *Disable* and the **Command Enable** switch to switch to *Home Mode*.
15. **Option 1:** If using the *virtual_QD2_MissionCtrl_2024a* Simulink model, use the joystick toggles to *Land* and then *Disarm*.



NOTE: Current release of joystick firmware should read "Quanser V2" which is compatible with the mapping in this model

16. **Option 2:** If using the *virtual_QD2_MissionCtrl_Keyboard_2024a* Simulink model, go to the Keyboard Command subsystem to *Land* and *Disarm*.
17. The QDrone 2 should land.

18. Stop all Simulink models.